

Independent combinations, mass spacing and blind-window comparisons

Standalone study v5.8.4

18 September 2026

The 1 MeV study retains the nominal GP sources of v5.8.2–3 and compares blind half-widths 2.25, 2.4, 2.5 and 2.6 times the mass resolution. It refits the three experiments and their shared-coupling combination, and adds Fisher and equal-weight signed Stouffer combinations of the independent experimental responses at the same tested mass.

The independent combinations emphasize a coherent upward response near 92 MeV. At the baseline width, Fisher gives local/global $Z = 3.72/2.28$ over 19–250 MeV, and signed Stouffer gives 3.93/2.58. The common-coupling maximum remains at 66 MeV. These methods test different alternatives; their difference is explained by the coupling information and separate fitted amplitudes, not by omitting the mass-search correction.

Search / combination	m [MeV]	Local Z	Global p	Global Z
2015 full	51	3.359	0.0194	2.067
2016 full	91	2.823	0.1163	1.194
2021 10%	66	2.489	0.3670	0.340
Shared coupling	66	2.829	0.2085	0.812
Fisher (atom calibrated)	92	3.723	0.0113	2.280
Equal-weight signed Stouffer	92	3.934	0.0049	2.580

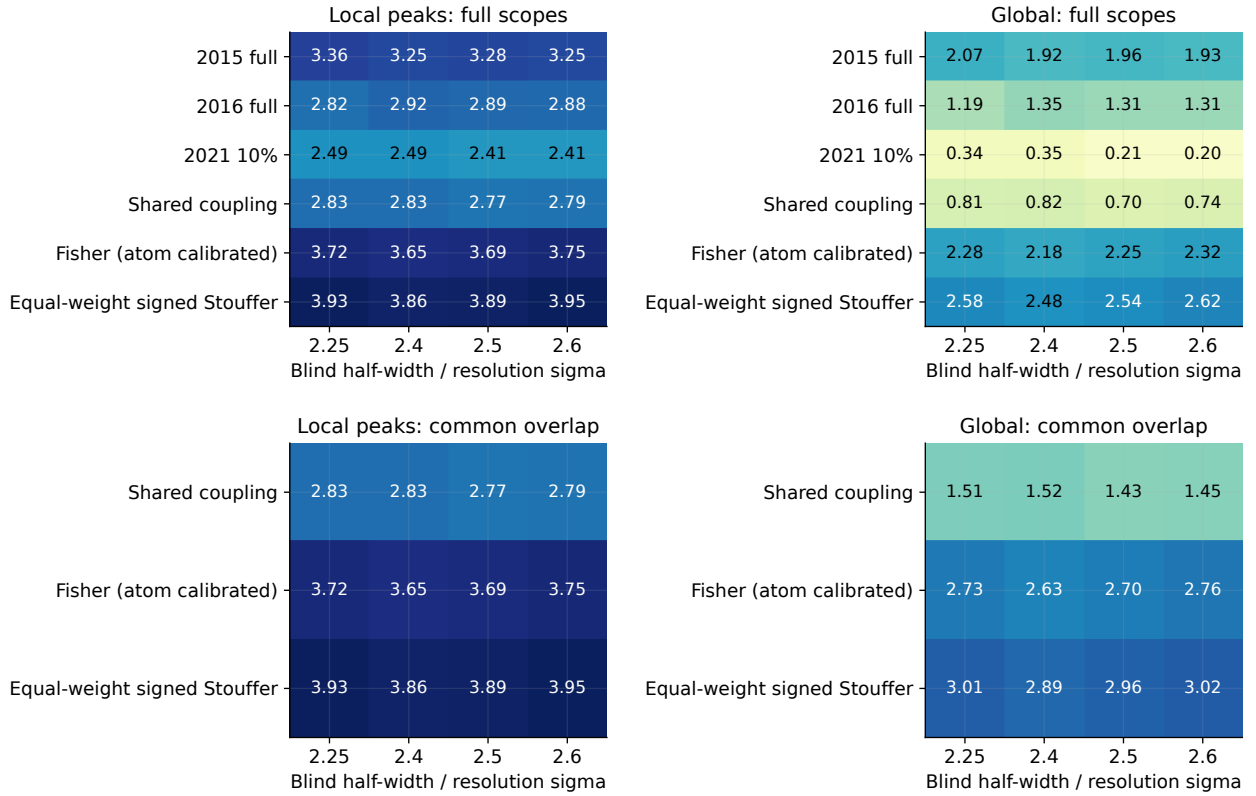


Figure 1. Peak local and global significances for every declared width. “Full” means each experiment’s own range, or the combined active-dataset range 19–250 MeV. The common-overlap range is 50–100 MeV. These are conditional response-GP estimates; choosing the most favorable method or width is not calibrated by these scope-wise p-values.

Does 1 MeV spacing reduce the penalty?

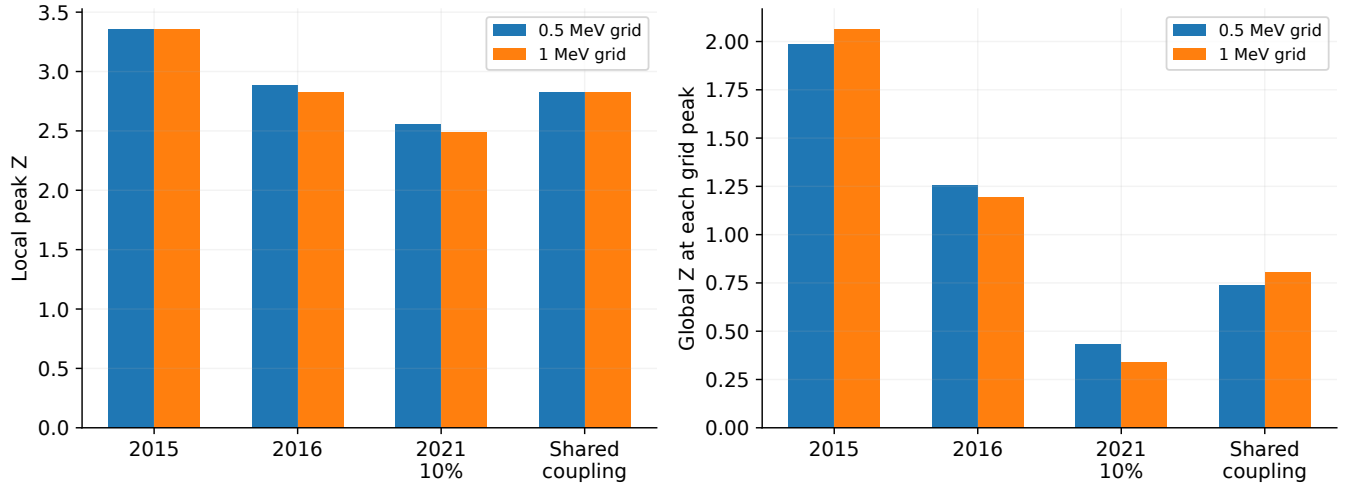


Figure 2. Nested 0.5 and 1 MeV baseline grids, using the same previously saved 200,000 Gaussian fields. The compared observed maximum is selected separately on each grid. This paired calculation isolates spacing from changes in source, fit, or random ensemble.

Scope	Peak masses	Local Z	Global Z	p at fixed $Z = 3$
2015 full	51 / 51	3.359 / 3.359	1.987 / 2.065	0.0701 / 0.0597
2016 full	91.5 / 91	2.886 / 2.823	1.254 / 1.194	0.0768 / 0.0715
2021 10%	65.5 / 66	2.558 / 2.489	0.430 / 0.341	0.1125 / 0.1055
Shared coupling	66 / 66	2.829 / 2.829	0.740 / 0.805	0.1482 / 0.1341

In the table each pair is 0.5 MeV / 1 MeV. At a fixed threshold, a subset of sampled masses always has a maximum no larger than the finer grid. Its tail probability therefore falls. This does not guarantee a more significant observed search: the coarser grid can miss the observed peak as well.

For 2016 the finer grid reaches local $Z = 2.886$ at 91.5 MeV, whereas the integer grid reaches 2.823 at 91 MeV. The observed global significance decreases from 1.254 to 1.194 despite the smaller fixed-threshold penalty. The same effect occurs in 2021. The 2015 peak at 51 MeV and coupled peak at 66 MeV are retained, so their global significances increase slightly.

The new width/method comparisons use 100,000 fresh Gaussian fields per width and scope; small Monte Carlo differences from this paired 200,000-field table are expected. The main baseline coupled result, for example, is $Z = 0.812$ versus 0.805 in this paired spacing test.

The 1 MeV results describe a declared discrete search. They do not establish convergence to a continuous mass search. Adopting coarse sampling because it produces a smaller p-value would be a separate selection decision; the grid comparison here reports both outcomes rather than replacing the original result with whichever is larger.

Common mass with independent signal strengths

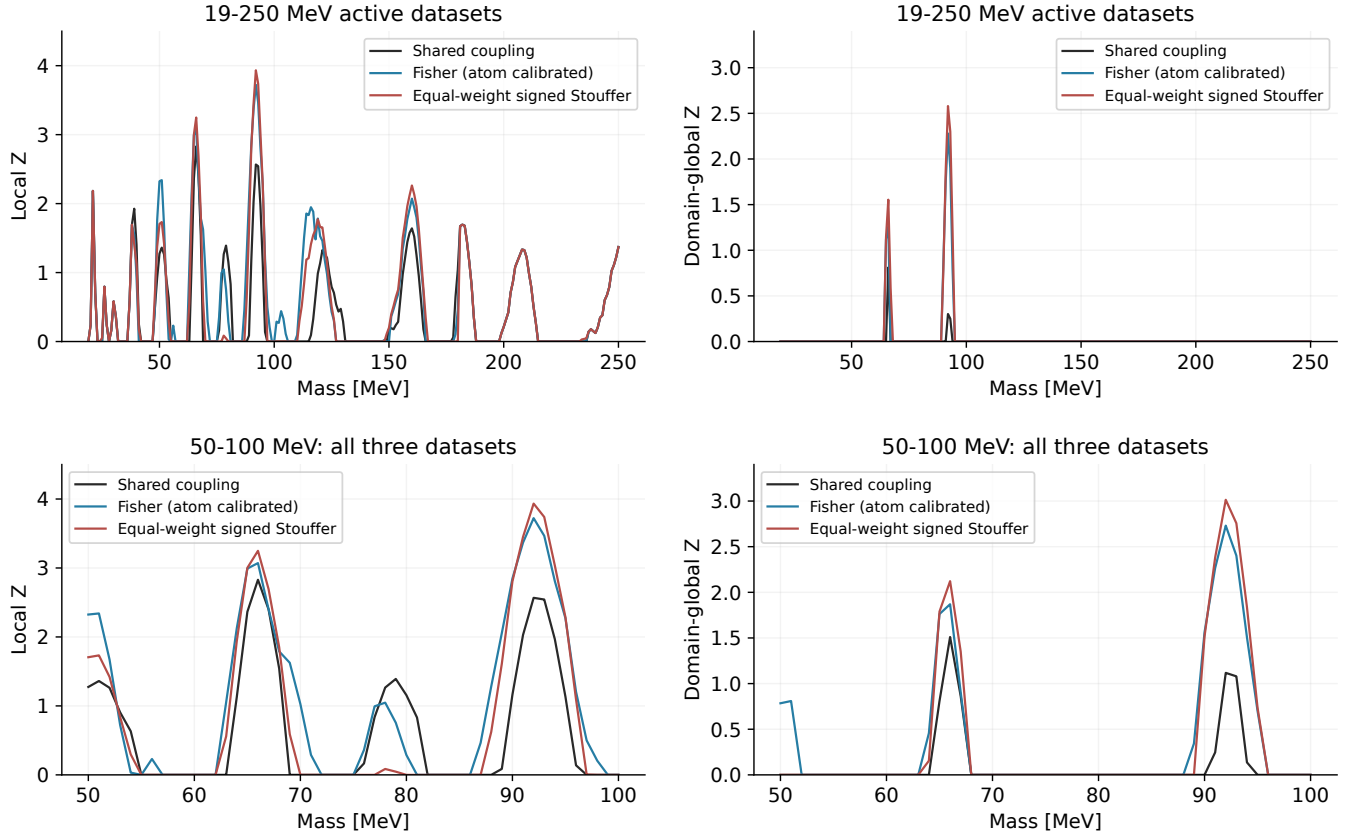


Figure 3. Baseline $\pm 2.25\sigma$ comparison. Fisher and signed Stouffer use the three independent experiments' responses at the same tested mass; they impose no common coupling or yield ratio. Their global curves still include the full indicated correlated mass search. All three datasets contribute throughout the lower-row overlap. Outside that overlap, the full-range curves use the datasets whose declared search ranges contain the mass.

The individual ranges remain 19–100 MeV for 2015, 39–180 MeV for 2016, and 50–250 MeV for 2021 10%. A change of participating datasets is included in each simulated combined scan. Nearby masses are correlated through their use of the same fluctuated spectra.

At 92 MeV the three reference-standardized scores are 2.594, 2.796 and 1.423. Their equal-weight sum divided by $\sqrt{3}$ is 3.934. The shared-coupling score at that same mass is 2.567, while its largest score remains 2.829 at 66 MeV. Thus the alternative combinations expose a different feature of the same spectra.

For the baseline 50–100 MeV overlap, Fisher gives global $Z = 2.73$ and signed Stouffer $Z = 3.01$. These are probabilities for a smaller, separately declared search domain. They neither include choosing between methods after inspection nor replace the full-range results.

Fisher calibration and the meaning of the alternative tests

For each experiment, the fixed-source response model is

$$r_i^*(m) = a_i(m) + s_i(m)W_i(m), \quad W_i \sim N(0, R_i), \quad z_i = \frac{r_{i,\text{obs}} - a_i}{s_i}.$$

The W_i fields are independent across experiments and correlated across mass within each experiment. The background GP is retrained on each Poisson spectrum's sidebands; the full generating GP source is held fixed.

The existing excess convention assigns $p_i = \bar{\Phi}(z_i)$ when $r_{i,\text{obs}} > 0$, and $p_i = 1$ otherwise. Fisher's statistic is $T = -2 \sum_i \log p_i$. The usual χ^2_{2k} formula presumes continuous uniform null p-values [1]. Here the p-values have an atom at one, so that conversion is generally conservative rather than the exact local tail.

An exact mixture within the Gaussian response approximation accounts for the atom. Put $q_i = \Phi(a_i/s_i)$. Under the null, $p_i = U_i$ for $U_i < q_i$, otherwise $p_i = 1$, with independent $U_i \sim \text{Uniform}(0, 1)$. For the subset J of positive fits,

$$w_J = \prod_{i \in J} q_i \prod_{i \notin J} (1 - q_i), \quad T \mid J = -2 \sum_{i \in J} \log q_i + \chi^2_{2|J|}.$$

For observed $T > 0$ we sum the corresponding tails over nonempty J ; at $T = 0$ the inclusive tail is one. The implementation recovers the ordinary Fisher formula when all $q_i = 1$, and reproduces the original local test when only one dataset contributes. The uncorrected χ^2 conversion is also retained in the ledger for comparison.

Signed Stouffer instead uses $Z_S = \sum_i z_i / \sqrt{k}$ and $p_S = \bar{\Phi}(Z_S)$. It combines continuous signed reference scores, so deficits reduce the sum. It has no requirement that every raw fitted signal yield be positive. It is a different, explicitly specified upward-response ordering. Each Gaussian scan repeats these combinations and records the largest $-\log p_F$ or Z_S over the stated domain. No scan-point independence approximation is used.

Combining experiment-global p-values answers another question

One may also apply ordinary Fisher to the three individual *global* p-value estimates. This permits a different peak mass in each experiment, so it tests for unusual behavior somewhere in the independent searches, not a common resonance mass. No additional mass penalty is required beyond those already included in the inputs. Its numerical conversion is approximate because the inputs are finite-Monte-Carlo estimates.

Half-width	Individual peak masses [MeV]	Fisher p	Z
2.25	51,91,66	0.0275	1.918
2.4	50,91,66	0.0290	1.895
2.5	51,91,66	0.0314	1.860
2.6	51,91,66	0.0337	1.829

The common-mass scans in Figure 3 and this table must not be interpreted as equivalent evidence for the same particle hypothesis.

Why the shared coupling suppresses the 92 MeV response

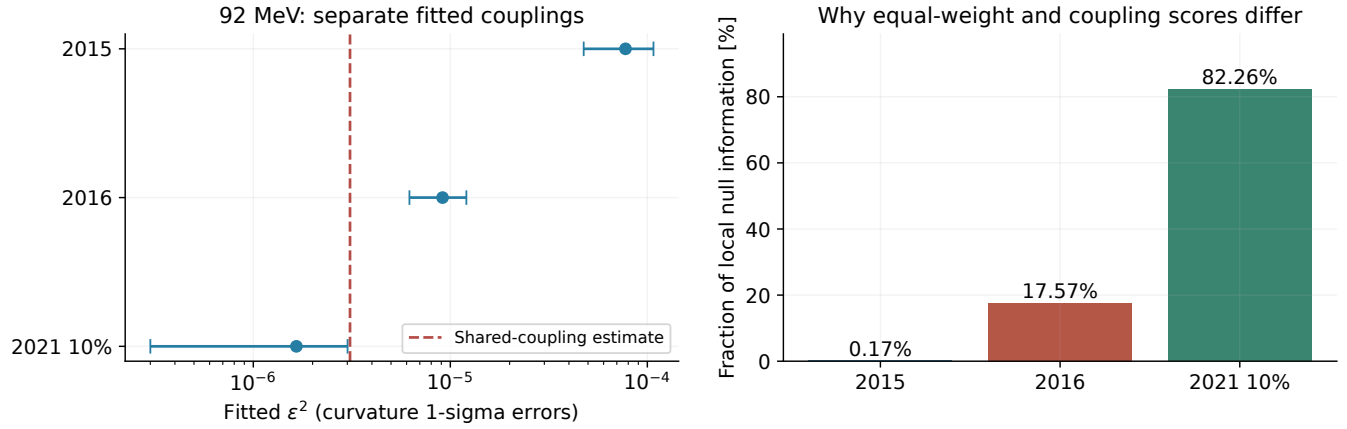


Figure 4. Baseline fits at 92 MeV. Left: separate coupling estimates with conditional likelihood-curvature errors; dashed line: the shared estimate. Right: fractions of local null Fisher information in that common-coupling parameter. These weights are not event fractions or independent measurements of detector efficiency.

The separate estimates are approximately 7.75×10^{-5} , 9.13×10^{-6} and 1.65×10^{-6} in ϵ^2 , while the common estimate is 3.10×10^{-6} . In a local Gaussian approximation, a common-parameter score weights the standardized experimental scores by $\sqrt{I_i}/\sum I_i$: approximately 0.041, 0.419 and 0.907 here. Stouffer uses $1/\sqrt{3}$ for each. It therefore gives substantially more weight to the 2015 upward fluctuation.

The observed likelihood improvement from allowing three unconstrained coupling parameters instead of one is $2(\text{NLL}_{\text{shared}} - \sum_i \text{NLL}_{i,\text{free}}) = 11.55$ at 92 MeV. This is a fit-compatibility diagnostic at an inspected mass, not a separately calibrated discovery or incompatibility significance. A larger independent-combination score does not show that the prescribed common-coupling relation is erroneous.

Independent combinations across blind widths

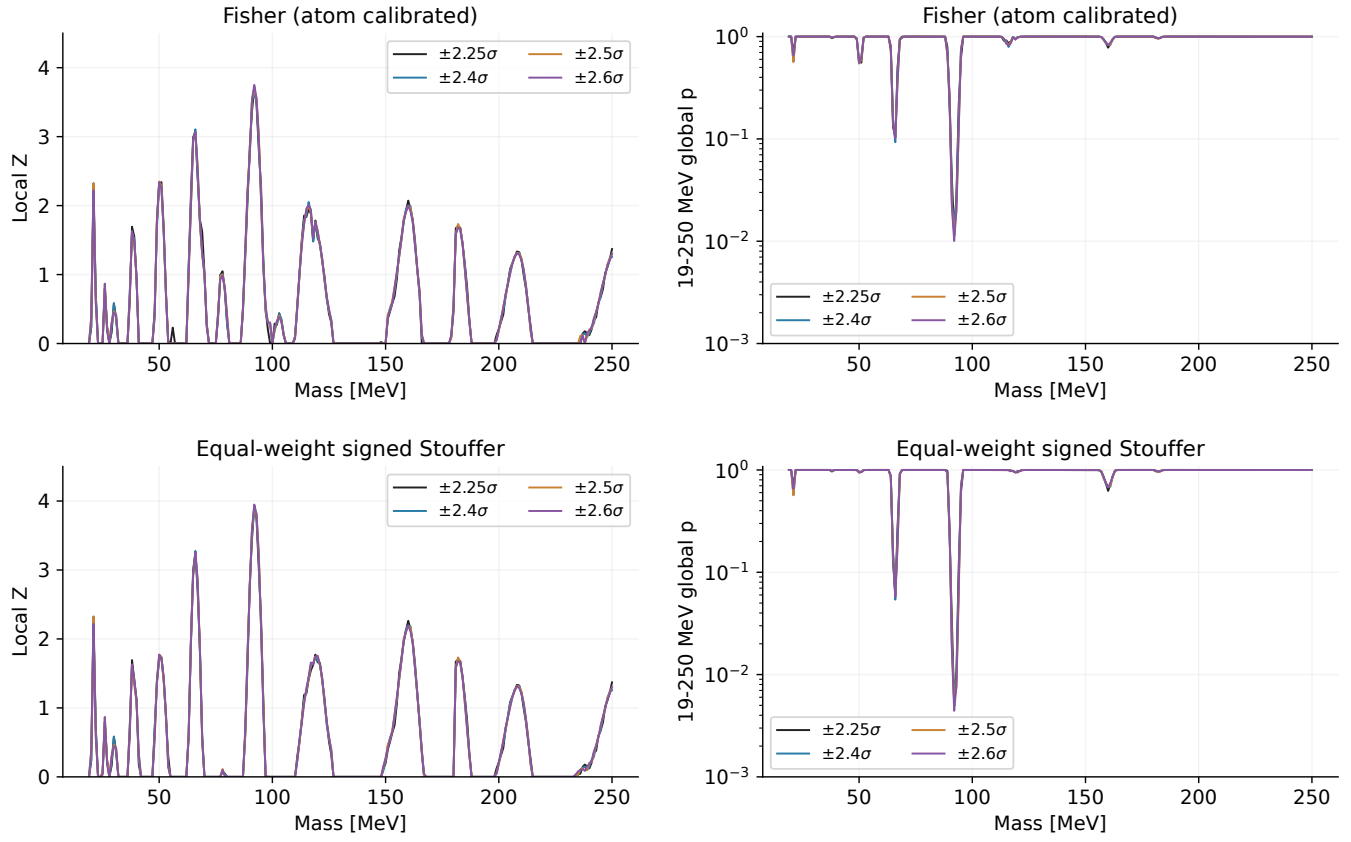


Figure 5. Alternative combinations across all blind widths: local Z and full-range global p . Their peaks remain at 92 MeV. Fisher full-range global Z spans 2.18–2.32 and signed Stouffer spans 2.48–2.62 across the declared widths.

The 92 MeV feature is stable under this particular change of window prescription. Its interpretation still depends on the tested signal relation and the fixed nominal background sources. The highest displayed result is not corrected for choosing a width or combination method.

Fisher accumulates evidence through small p -values and retains the original nonpositive-fit atom. Signed Stouffer uses all three signed reference responses, including deficits, with equal weights. Their numerical agreement on the peak location is useful, but it is not an independent replication: both statistics use the same experimental data and fitted responses.

Changing the blind half-width: local response

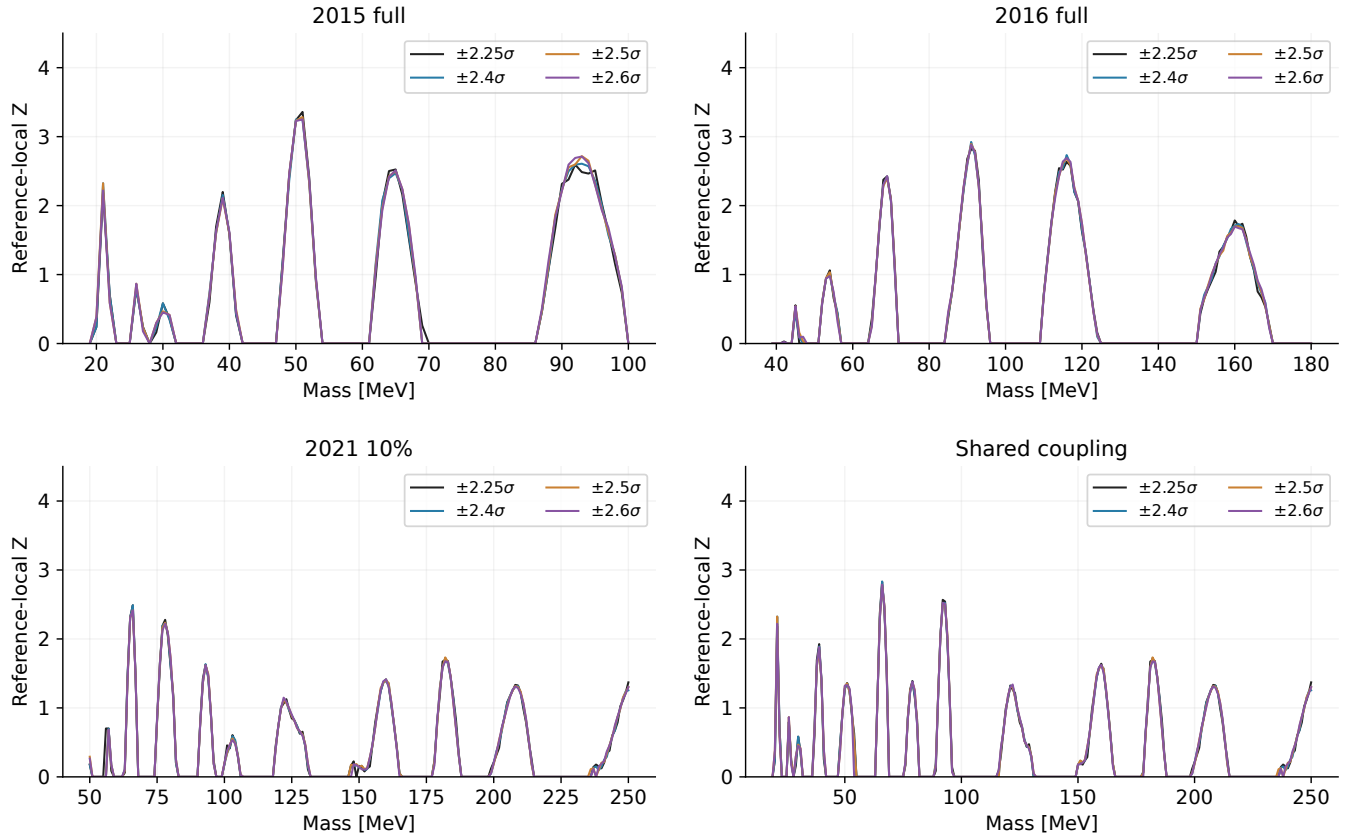


Figure 6. Reference-local significance for each experiment and the shared-coupling search. The half-width changes both the GP held-out region and the likelihood signal window, following the existing implementation. The source means, reviewed mass-dependent kernels, supports, resolution and signal normalization remain fixed. The full Gaussian signal template is restricted to each window without renormalizing away its missing tails.

Half-width	2015	2016	2021	Coupled	Fisher	Stouffer
2.25	2.067	1.194	0.340	0.812	2.280	2.580
2.4	1.916	1.351	0.353	0.821	2.182	2.484
2.5	1.962	1.309	0.211	0.704	2.252	2.541
2.6	1.926	1.308	0.204	0.736	2.325	2.618

The table reports global Z at each width's local maximum over the corresponding full scope. It is a comparison across declared configurations, not a width-profiled search. The covariance, offsets and response scales are recomputed for every width, so no old trials factor is simply attached to a newly fitted peak.

For 2016 the integer-grid local peak improves from 2.823 at $\pm 2.25\sigma$ to 2.923 at $\pm 2.4\sigma$, with more modest changes at the other widths. The response does not improve monotonically with window size. Some 2021 curves coincide at individual masses because a small width change need not add or remove an analysis bin there.

Wider blinding changes interpolation distance and background uncertainty as well as the fitted events. This is a controlled window comparison with frozen kernel settings; it is not a retuned background-model selection study.

Changing the blind half-width: global probabilities

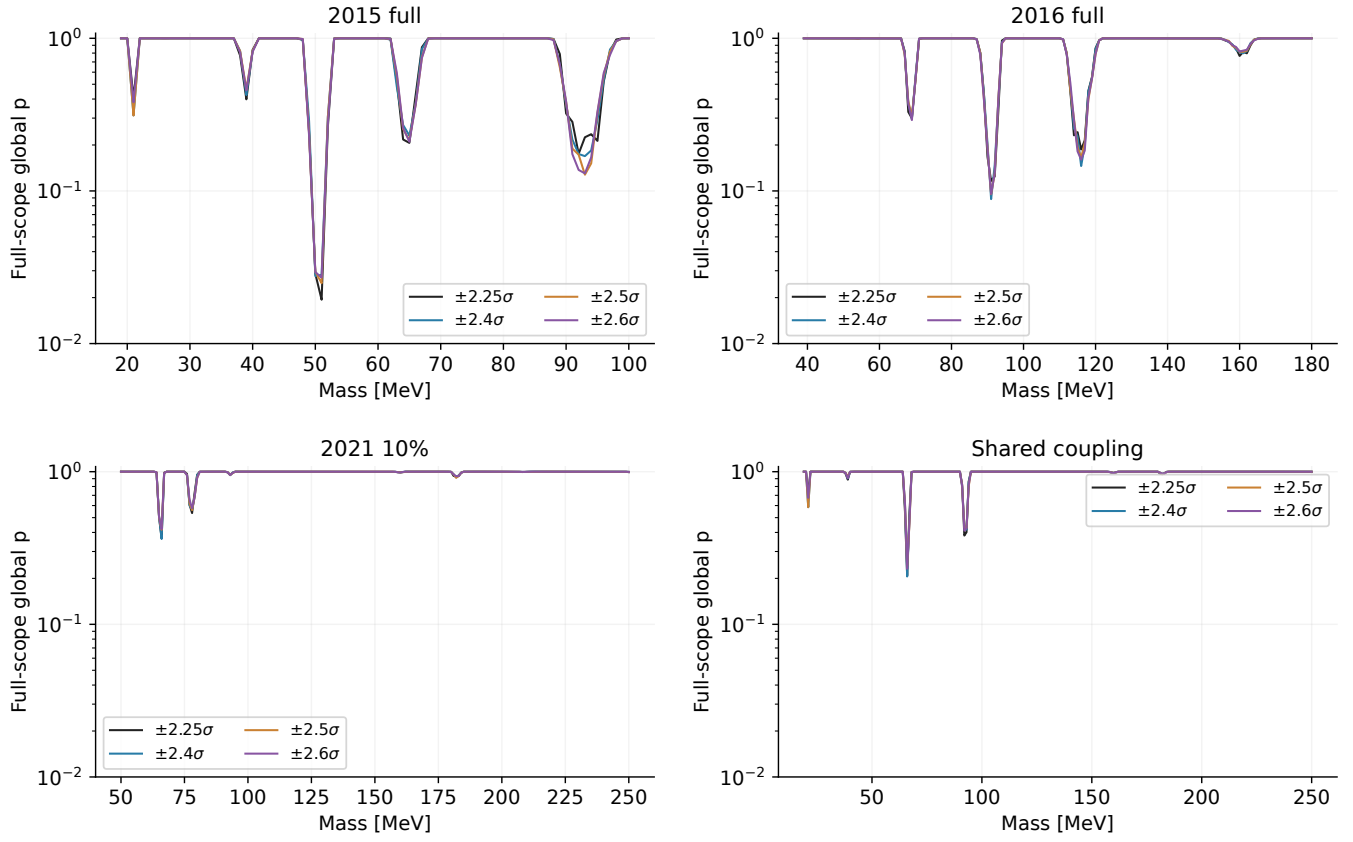


Figure 7. Full-scope global probabilities from 100,000 Gaussian fields for each configuration. Local and global numerical probabilities, positive one-sided Z , exceedance counts, exact binomial intervals and direct-Poisson counts are retained at every plotted mass. The global probability at any point compares that score with the largest score anywhere in the declared scope.

The shared-coupling peak remains near 66 MeV for every width and its full-range global significance spans roughly 0.70–0.82. Independent combinations instead emphasize 92 MeV, where the experiments all have upward reference-standardized scores but do not prefer the same fitted coupling value.

The width changes are small compared with the distinction between the common-coupling and independent-combination hypotheses. Wider masks also do not remove the deterministic source response: the 2016 RMS offset remains approximately 0.61–0.62 on the integer grid. This study therefore gives no evidence that simply widening the blind region resolves the source-model question discussed in v5.8.3.

All displayed probabilities condition on the nominal GP sources. The 2016 $\pm 2.6\sigma$ peak has a modest direct-tail validation discrepancy, detailed on the final page. The Gaussian curves are conditional approximations, not unconditional discovery calibrations.

Observed 90% CLs reach

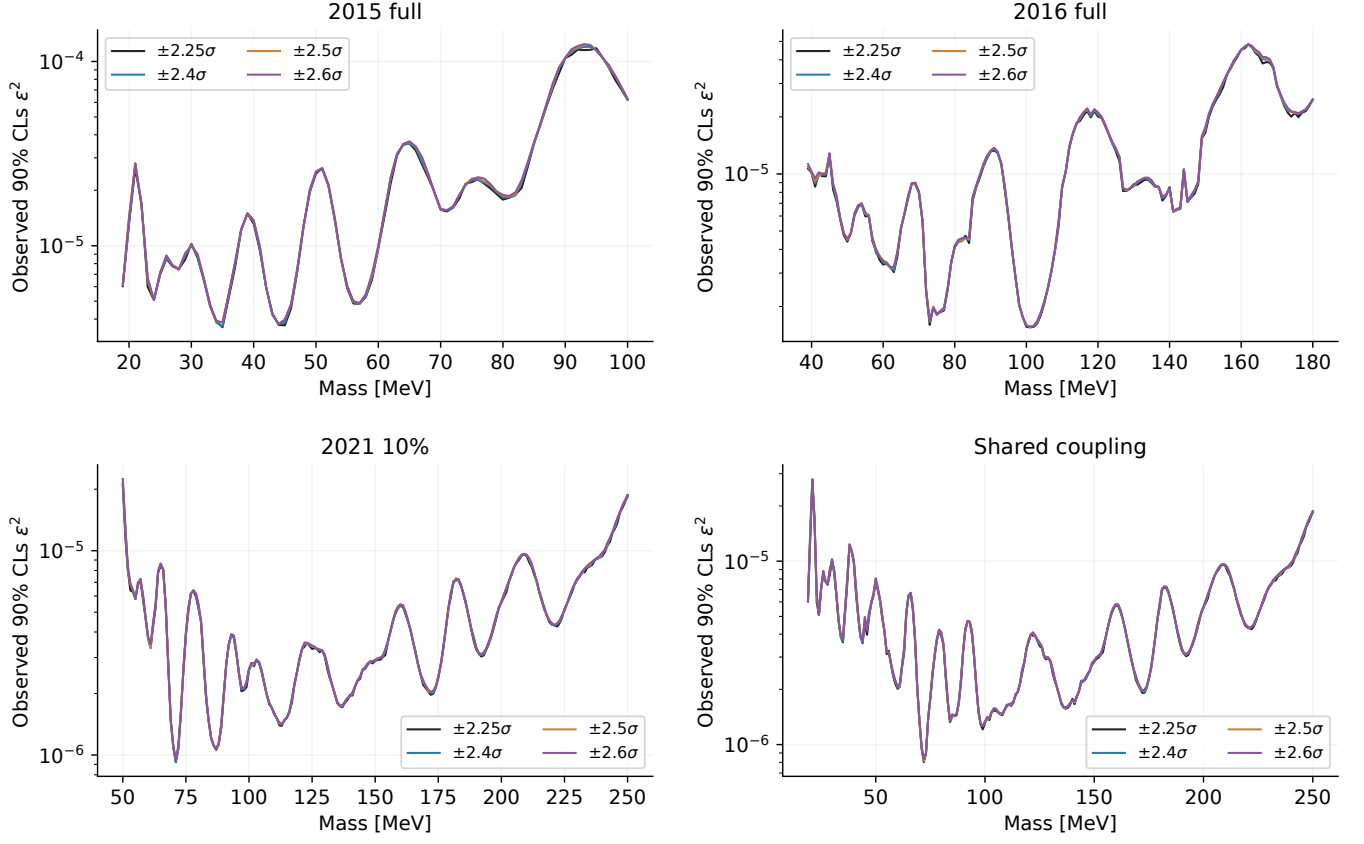


Figure 8. Recomputed observed 90% CLs upper endpoints in ϵ^2 , for the individual experiments and the common-coupling combination. Lower endpoints are stronger exclusions under the prescribed signal model. Every plotted endpoint is obtained from the bounded profile-likelihood CLs solver with its original signal-to-coupling conversion.

At every mass and width the solver profiles the Poisson likelihood with its correlated Gaussian background constraint, handles the physical-boundary/deficit branch, and solves $\text{CL}_s = 0.1$. It checks the root, likelihood nesting, positive Poisson means and optimizer convergence. The new Fisher and Stouffer significance calculations are not substituted into this upper-limit construction.

Fisher combination by itself does not specify how a common coupling predicts the separate event yields. It therefore supplies no unique one-dimensional ϵ^2 reach curve. Constructing such a curve would require a signal-strength model and its signal-plus-background calibration; imposing the existing physical yield relation would restore the common-coupling hypothesis. Separate experimental limits and the existing common-coupling limits are the well-defined reach comparisons here.

At $\pm 2.6\sigma$, the median observed upper-endpoint ratios to baseline are 1.031 for 2015, 1.022 for 2016, 1.017 for 2021 and 1.020 for the combination. Widening the window therefore slightly weakens the median observed reach in this comparison, even though individual masses can move in either direction.

Reach ratios and the model-Asimov comparison

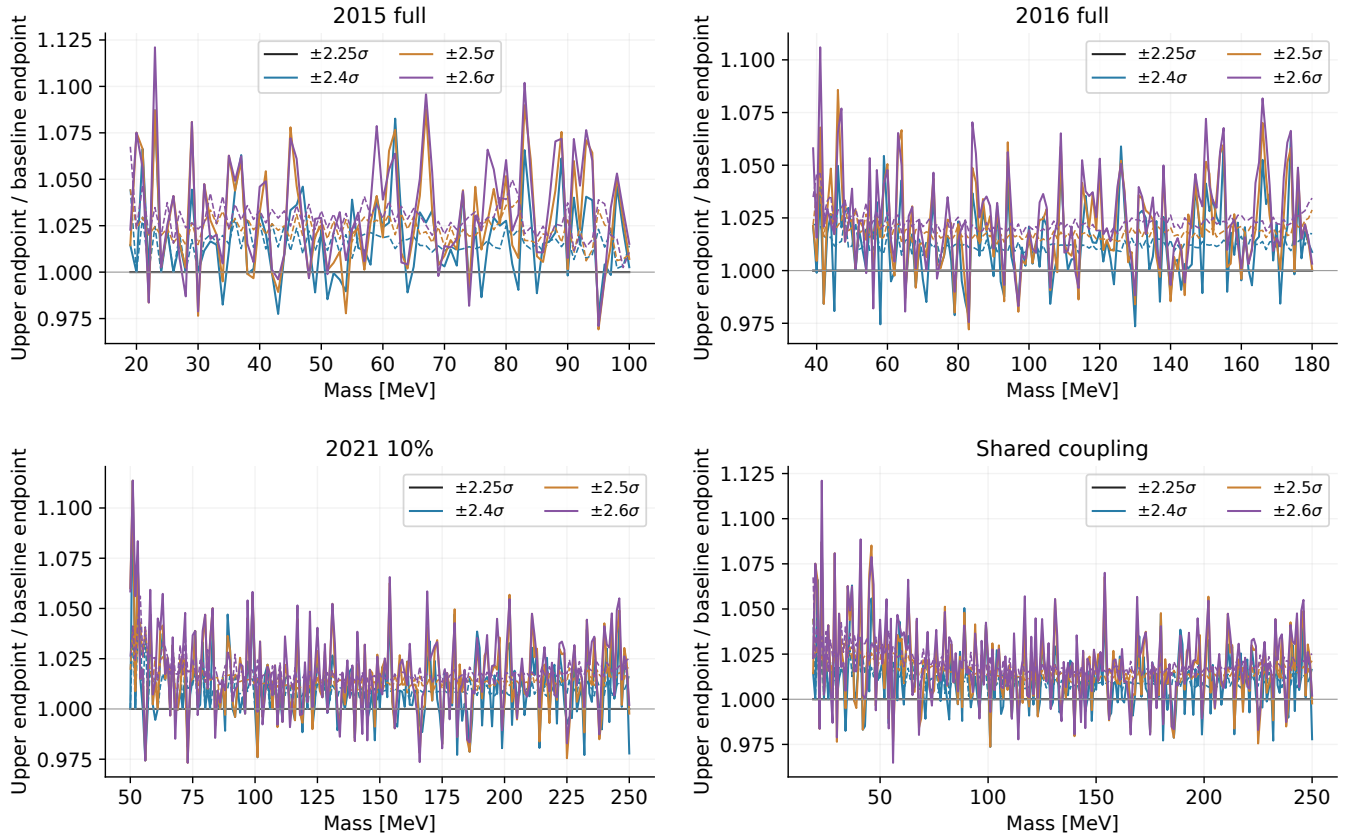


Figure 9. Upper endpoints divided by the $\pm 2.25\sigma$ result at the same mass. Solid: observed spectra. Dashed: a deterministic local model-background Asimov calculation. The black baseline is one. Ratios below one indicate a stronger endpoint; they are not estimates of statistical significance or discovery power.

The Asimov calculation sets the fitted-window counts to the predictive mean of that mass-local likelihood, with its GP nuisance covariance held fixed, then solves the same 90% CLs equation. It is a conditional model sensitivity reference. It is not a coherent full-spectrum experiment, a median proven by an ensemble, an expected band, or an added fluctuated toy sample.

At $\pm 2.6\sigma$, the median model-Asimov endpoint increases by 2.77%, 2.26%, 1.73% and 1.81% for 2015, 2016, 2021 and the combination. The combined observed ratio spans 0.965–1.121, whereas its Asimov ratio spans 1.008–1.068. The oscillating observed changes thus include the particular fluctuations moved into or out of the fit, in addition to the loss of interpolation information from a wider exclusion region.

Absolute model-Asimov curves and an additional global- Z width comparison are included as separate vector plots. All extrema and their mass locations are recorded in `results/reach_ratios_summary.csv`. These results do not select a new production window or establish coverage for any of the alternatives.

Validation, interpretation and reproducibility

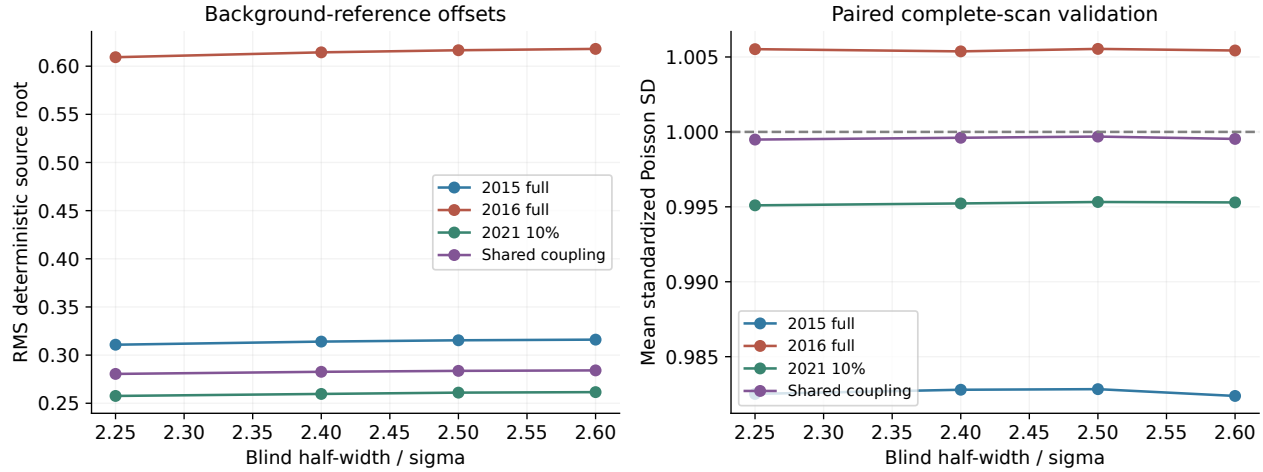


Figure 10. Deterministic source-offset RMS and mean standardized widths from 256 paired complete Poisson scans. The same full-spectrum realization is used at every mass and at every blind width; independent dataset draws are combined by matching realization index. These repeated configurations are therefore correlated, not additional independent experiments.

The numerical scan contains 928 width/mass checkpoint pairs, 2,628 fitted scope/mass configurations and 5,096 reported scope/domain/mass rows after the alternative combinations and overlap views are added. All 124 numerical/provenance checks pass. The largest scalar/batch root discrepancy is 5.27×10^{-10} ; baseline replay differs from the frozen roots by at most 1.32×10^{-6} across numerical runtimes. The width-dependent derivatives pass 264 finite-difference checks with maximum discrepancy 2.94×10^{-6} . Numerical covariance-conditioning derivatives remain omitted, as in the prior validated approximation.

One statistical tail check is flagged. For full 2016 at $\pm 2.6\sigma$, the Gaussian peak-global probability is 0.09547. Direct complete Poisson scans give 35/256, with a two-sided 95% interval 0.0971–0.1850. The Gaussian estimate lies just below this interval. Its direct exceedance rate at the Gaussian 95th-percentile maximum is 14/256, consistent with 5%. This single comparison among correlated checks is not conclusive evidence of a systematic failure, but more complete scans are needed to separate a modest tail mismatch from sampling variation. It is retained rather than treated as an automatic pass.

All other reported Gaussian peak probabilities fall within their corresponding direct 95% intervals. The most interesting independent-combination tails remain sparsely checked: baseline full-range Fisher has 4/256 direct exceedances, signed Stouffer 1/256, and overlap Stouffer 0/256. Zero does not mean zero probability; its one-sided 95% upper bound is about 0.0116. The Gaussian estimates near 3σ in the overlap are not precise direct-Poisson measurements.

The full source-building step is held fixed. Source-estimation uncertainty, signal absorption during source construction, correlated systematic uncertainties beyond the specified conditional model, and selection among widths, methods and scopes are not calibrated here. These are the interpretation limits of the reported results.

The package contains pinned inputs, full fitting and analysis scripts, checkpoints, response fields, numerical ledgers and render QA. The README gives restart and rebuild commands. Prior releases are unchanged.

References

1. SciPy, [combine_pvalues](#) documentation: Fisher and Stouffer constructions and the continuous independent-uniform assumption. The atom mixture above is derived for this study’s stated convention.
2. V. Ananiev and A. L. Read, *Gaussian Process-based calculation of look-elsewhere trials factor*, [arXiv:2206.12328](#).
3. G. Cowan et al., *Asymptotic formulae for likelihood-based tests of new physics*, [arXiv:1007.1727](#); HPS-GPR v5.0.5 and studies v5.8.0–3 for the frozen likelihood, GP-source and CLs conventions.

HPS Gaussian-Process Resonance Search

Common-mass likelihood and mass-coherence study
Analysis Note v5.8.5

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Abstract

We examine the structure near 92 MeV in the HPS 2015, 2016, and 2021 (10%) samples using a common-mass resonance likelihood with independent nonnegative signal amplitudes. The scan covers 19–250 MeV in 1 MeV steps, with each dataset included within its declared support. For the nominal blind half-width of $2.25\sigma_m$, the combined maximum occurs at 92 MeV. The conditional Gaussian-response calibration gives local and full-domain significances of 3.583 and 2.060, respectively. Seven of 256 direct Poisson scans exceed the observed maximum, giving a 95% probability interval of $[0.0111, 0.0555]$. Across four blind widths, the combined maximum remains at 91–92 MeV. Individual peak stability is less uniform: the 2021 diagnostic maximum switches from 93 MeV to a competing excursion at 80 MeV. A toy-calibrated mass-coherence statistic gives $p = 0.763$, showing no unusual clustering under the specified null. These probabilities condition on the pinned GP background source and exclude corrections for choices made after inspecting the data.

1 Introduction

This continuation tests a resonance at one common mass with independent nonnegative signal amplitudes in the 2015, 2016 and 2021 (10%) samples. It reuses the pinned v5.8.4 spectra, likelihoods, background sources, response derivatives and 256 complete Poisson scans. The full search remains the integer grid 19–250 MeV, with each dataset included only on its inherited support.

Method	Peak [MeV]	Local Z	Global p	Global Z	Direct k/N
Shared coupling	66	2.83	0.2085	0.81	41/256
Fisher	92	3.72	0.0113	2.28	4/256
signed Stouffer	92	3.93	0.0049	2.58	1/256
Free amplitudes	92	3.58	0.0197	2.06	7/256

Table 1: Full-domain peaks at $\pm 2.25\sigma_m$. Local and global Z use the conditional Gaussian-response model. Direct counts use the same 256 full Poisson realizations. Old methods are unchanged v5.8.4 results; the new test has its own calibrated null distribution.

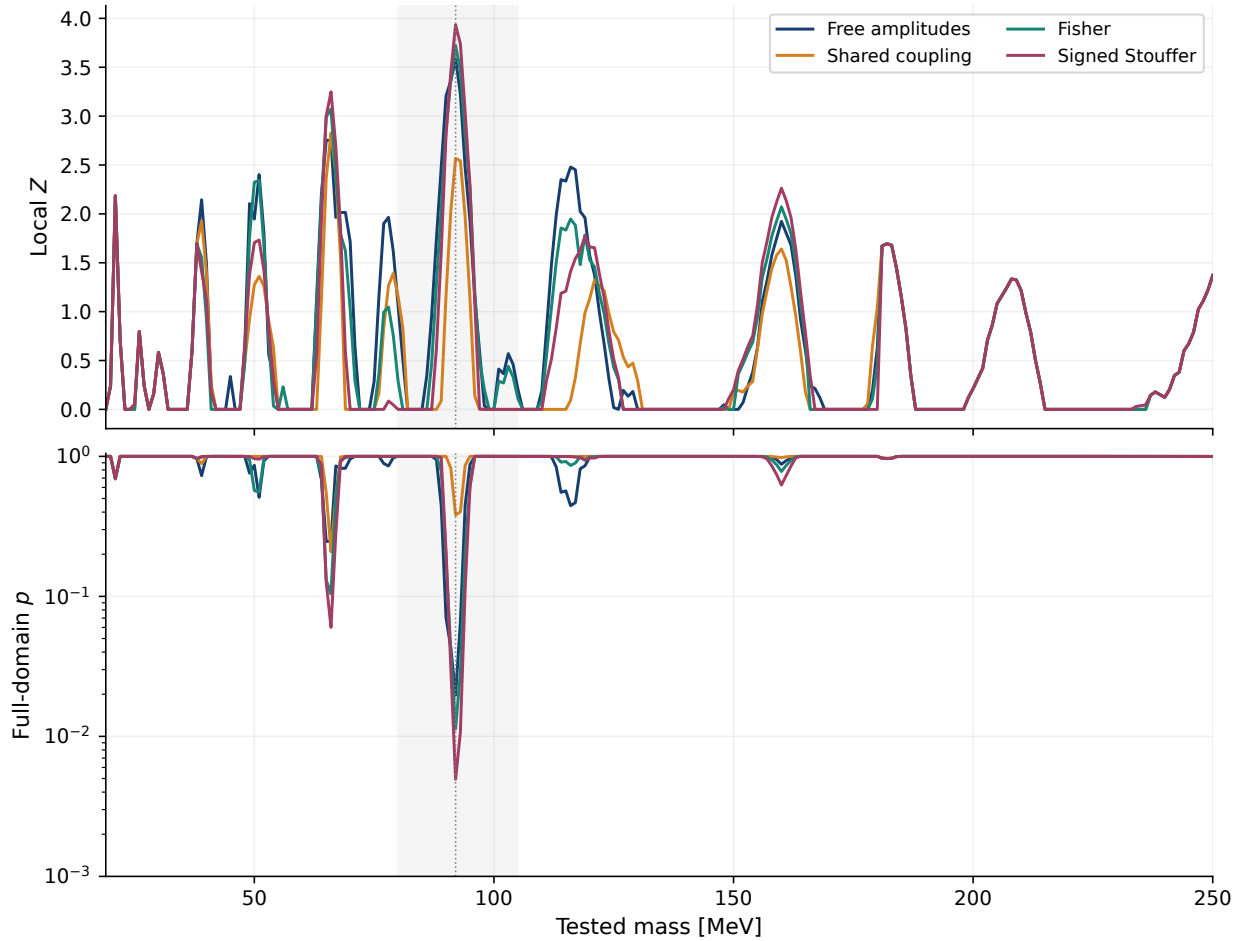


Figure 1: Baseline $\pm 2.25\sigma_m$ comparison: local significance and probability for an equally strong score anywhere on the full 19–250 MeV grid. Curves use conditional Gaussian-response calibration; Table 1 retains the direct Poisson count for the new test. Shading marks the post-hoc 80–105 MeV diagnostic interval. Each method uses its own declared ordering and null maxima.

2 Local and domain-global probabilities

At a mass fixed independently of the data, the local probability is

$$p_{\text{loc}}(m) = \Pr_0\{T(m) \geq T_{\text{obs}}(m)\}, \quad Z_{\text{loc}}(m) = \Phi^{-1}[1 - p_{\text{loc}}(m)], \quad (1)$$

where larger T is more signal-like and the probability includes equality. This inclusive convention matters when a nonnegative fitted amplitude gives a point mass at $T = 0$: its inclusive tail is one. A displayed nonnegative Z is a plotting convention; the probability remains the primary quantity. The common-mass alternative has one amplitude for each active dataset, so $\sqrt{q_R}$ is not generally its local significance. Its null distribution must be calibrated with the appropriate boundary and active dataset count.

For a declared mass grid $\mathcal{G}$ and a declared local ordering $S(m)$, the scan statistic and its domain-global probability are

$$M = \max_{m \in \mathcal{G}} S(m), \quad p_{\mathcal{G}} = \Pr_0\{M \geq M_{\text{obs}}\}. \quad (2)$$

For example, $S = -\log p_{\text{loc}}$ compares locally calibrated evidence across masses; $S = q_R$ defines a different valid scan test. The observed maximum and every null maximum must use the same ordering. A full-spectrum toy supplies the entire correlated curve, rather than independent random fluctuations at each mass. This accounts for the look-elsewhere effect within that grid. See Gross and Vitells for scan inference and Ananiev and Read for correlated Gaussian significance-field approximations.

Here the inherited full combined grid is 19, 20, $\dots$, 250 MeV. The active supports are 19–100 MeV (2015), 39–180 MeV (2016), and 50–250 MeV (2021, 10%); only 50–100 MeV contains all three datasets. “Common mass” ties the mass of the active components and does not extend an experiment’s support. Adding sites to a domain cannot reduce its null exceedance probability at a *fixed threshold*. Comparisons of observed maxima can also change the threshold. The number of grid points is not the number of independent trials: nearby masses share events, background fits, and signal templates. Finer sampling resolves maxima until grid convergence; a 1-MeV-grid calibration is not, by itself, a continuous-mass calibration.

The 80–105 MeV interval was chosen after inspection and is used only to describe the reported structure. Its 2015 interval ends at 100 MeV. Calibrating a statistic restricted to this interval answers a conditional diagnostic question; it does not replace the full-domain discovery test or correct the choice of this interval. Likewise, four width-specific global probabilities are four conditional results, not a selection-corrected claim based on the most favorable width. Such a claim would require repeating the entire width/method/domain selection in joint null realizations. Reusing the same fluctuations across widths preserves their strong dependence.

All probabilities here condition on the pinned observed-data-derived GP source, its kernel policy, and the stated likelihood construction. Direct Poisson scans repeat the local fits; Gaussian response fields approximate their fluctuations. Neither propagates uncertainty in choosing or estimating the source itself. Exact likelihood factorization therefore does not imply unconditional physical-discovery calibration. Boundary asymptotics are reviewed by Cowan et al.; sparse toy tails here are reported with exceedance counts and binomial bounds.

3 Common-mass likelihood

For independent datasets, at fixed mass and width,

$$L(\boldsymbol{\mu}, \boldsymbol{\theta}; m, w) = \prod_{d \in A(m)} L_d(\mu_d, \theta_d; m, w), \quad \mu_d \geq 0.$$

Each local likelihood is the archived Poisson model with correlated Gaussian GP-background constraint. Its expected bin counts have the form $\lambda_d = b_d + L_d \theta_d + A_d S_d$ and penalty $\theta_d^T \theta_d / 2$. The engine uses the original template normalization A_d , with fitted-window yield $\mu_d = A_d \sum_i S_{di}$. This is a one-to-one amplitude reparameterization. Here L_d in the count equation is the covariance factor, distinct from the likelihood symbol. The local background prediction and covariance are conditioned on the sidebands of each observed or toy spectrum. The full generating source is fixed.

$$q_R(m; w) = 2 \left[\sum_d \text{NLL}_d(0, \hat{\theta}_{d,0}) - \sum_d \text{NLL}_d(\hat{\mu}_d^+, \hat{\theta}_d) \right] = \sum_{d \in A(m)} [\max(r_d(m; w), 0)]^2.$$

The archived signed root r_d uses the unconstrained single-amplitude fit and the zero-amplitude profile. Convexity and factorization make the clipped-root identity exact for this likelihood, including boundary fits. This is a direct likelihood construction; Fisher or Stouffer p-values do not enter q_R .

Independent amplitudes impose no common coupling or calibrated relative yield. They also allow one or two amplitudes to vanish, so rejection of the all-zero null does not establish a signal in every dataset.

The exact likelihood statistic must be distinguished from its sampling calibration. For the conditional response approximation,

$$r_d^*(m, w) = a_d(m, w) + D_{d,m,w}^T \xi_d, \quad \xi_d \sim N(0, I), \quad s_d^2 = D_{d,m,w}^T D_{d,m,w}.$$

Different datasets fluctuate independently. Within each dataset the same latent fluctuation drives the entire mass scan. The probability of $q_R = 0$ is $\prod_{d \in A(m)} \Phi[-a_d(m, w)/s_d(m, w)]$; its inclusive tail is one. The local tail of the sum of clipped squared normals is integrated over all nonempty positive-amplitude subsets with deterministic polar quadrature. At $a_d = 0, s_d = 1$ it reduces to the independent-boundary mixture $2^{-k} \sum_{j=1}^k \binom{k}{j} \Pr(\chi_j^2 \geq q)$ for $q > 0$, not a single χ_k^2 tail. Actual mass-dependent a_d, s_d are retained. Monotone lookup tables accelerate the scans. The declared ordering is $S_R(m) = -\log p_R(m)$.

The 256 stored Poisson realizations give an additional direct local and global calibration of this ordering, using the exact profile roots at every mass. At 92 MeV none exceeds the observed local statistic for any width; the one-sided 95% upper bound is $p < 0.01163$, which cannot resolve the response-model estimate near 3.6σ . Zero counts are never presented as zero probability. Point estimates elsewhere use $(k+1)/(N+1)$; two-sided 95% intervals are Clopper–Pearson. A separate 100,000-draw response ensemble preserves mass and width correlations through the joint derivative covariance.

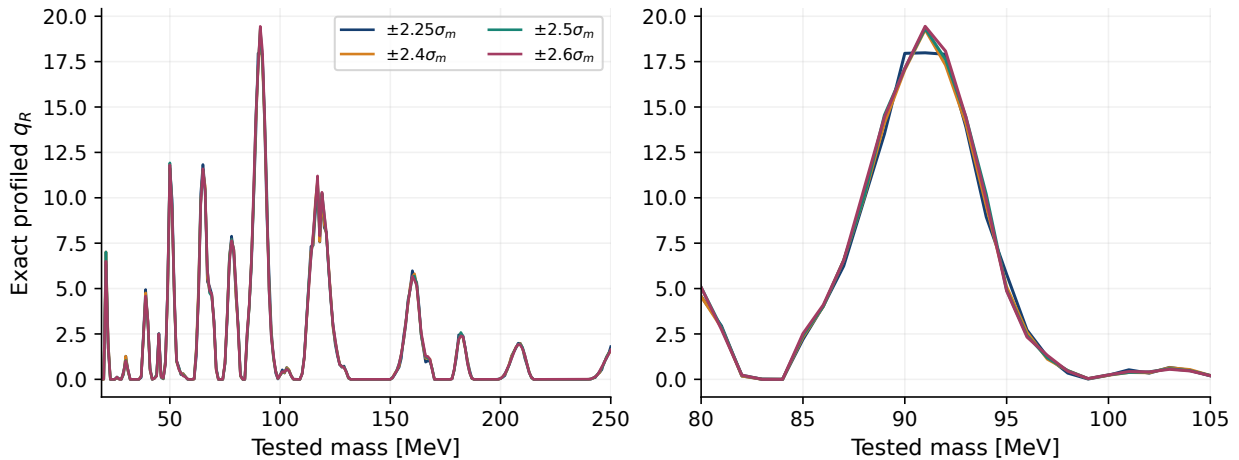


Figure 2: The exact observed profile statistic q_R for all four widths. Left: full domain, including changes in active datasets. Right: diagnostic interval. The global ordering uses calibrated $-\log p_R$, whose maximum can differ from the maximum of raw q_R . Neither ordinate is replaced by $\sqrt{q_R}$ as a significance.

4 Results of the combined search

At 92 MeV, $q_R = 17.307\text{--}18.077$, local $Z = 3.510\text{--}3.610$, and full-domain Gaussian-response $Z = 1.949\text{--}2.104$. At each width's selected peak, the direct global plus-one probabilities are 0.0311, 0.0350, 0.0272, 0.0272, corresponding to Z values 1.864, 1.812, 1.923, 1.923. Their intervals are much wider than the 100,000-field Monte Carlo errors. The new test is less significant than signed Stouffer: it tests a broader alternative with separately nonnegative amplitudes, and its

w	Peak [MeV]	Local Z	$Z(92)$	Global Z	$k/256$	Direct global p : 95% CI
2.25	92	3.583	3.583	2.060	7	[0.0111, 0.0555]
2.4	91	3.518	3.510	1.960	8	[0.0136, 0.0606]
2.5	92	3.551	3.551	2.017	6	[0.0086, 0.0503]
2.6	92	3.610	3.610	2.104	6	[0.0086, 0.0503]

Table 2: Free-amplitude peaks selected by maximum $-\log p_R$ on the full grid. The direct count column gives k out of 256. Significance columns are Gaussian-response estimates. Raw q_R instead has its maximum at 91 MeV for all four widths.

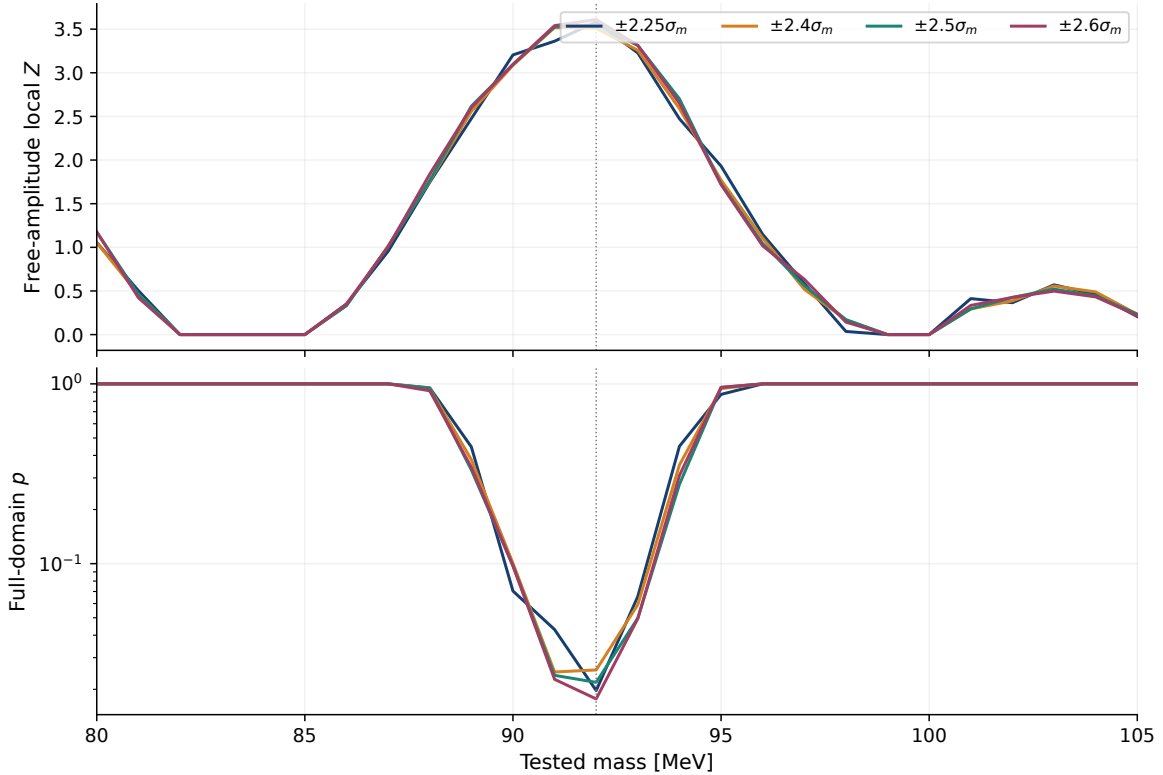


Figure 3: Free-amplitude local significance and full-domain probability, shown near the inspected structure. Every global probability in this zoom still uses the complete 19–250 MeV scan. Gaussian-response estimates are accompanied by direct Poisson counts and intervals in Table 2 and the numerical ledger. The four curves reuse the same observations and are not four independent confirmations.

own boundary-aware local calibration. The same width controls both the bins excluded from GP conditioning and the fitted likelihood window. Signal templates retain their original normalization. These are fixed-kernel window comparisons; no width has been adopted by choosing its largest observed significance. The displayed mass-global probabilities do not correct the post-observation comparison of widths or methods.

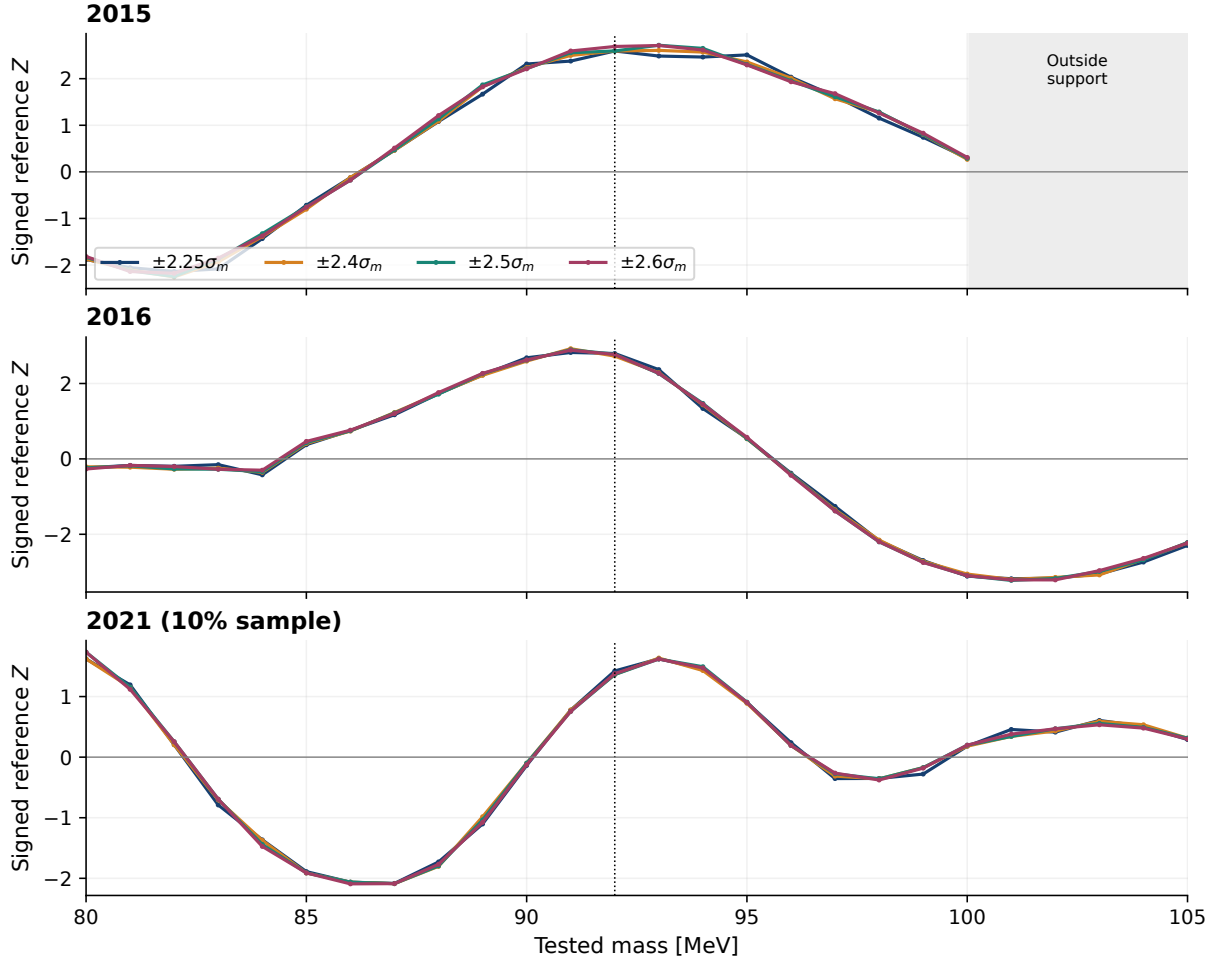


Figure 4: All twelve signed reference-score curves $Z_{d,w}(m) = (r_{d,w} - a_{d,w})/s_{d,w}$ in the post-hoc diagnostic interval. Dots mark the 1 MeV grid; the dotted vertical line is 92 MeV. Gray space beyond 100 MeV in the 2015 panel is outside its support. Signed scores retain deficits, unlike nonnegative significance displays.

Dataset	2.25	2.4	2.5	2.6	$\sigma(92)$	Δm	R
2015	92	93	93	93	4.804	1	0.208
2016	91	91	91	91	3.788	0	0.000
2021	93	93	80	80	2.449	13	5.309

Table 3: Diagnostic-region signed-score argmax masses [MeV] at each half-width; resolution and range are also in MeV. $\Delta m = \max_w \hat{m} - \min_w \hat{m}$ and $R = \Delta m/\sigma(92)$. The region is 80–105 MeV, clipped to 80–100 for 2015. All selected observed peaks also pass the positive-raw-fit gate used by the coherence test.

5 Mass stability across blind widths

The 2015 and 2016 maxima remain within one grid bin across width. In 2021 the first two widths have almost tied excursions at 80 and 93 MeV: at $w = 2.25$, their signed scores are 1.6216 and 1.6228. At $w = 2.5$ they are 1.7332 and 1.6194. Thus the 13 MeV range is an argmax rank switch involving the lower diagnostic boundary, not evidence that a tracked resonance shifts by 13 MeV. The response at 92 MeV remains positive but modest.

5.1 Individual amplitudes at 92 MeV

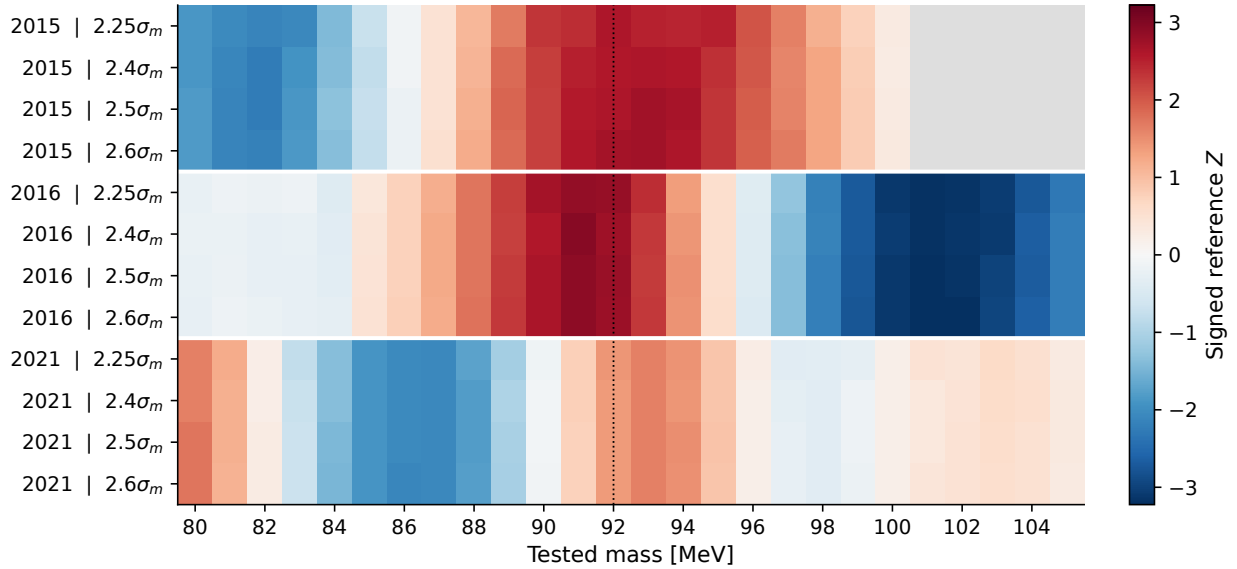


Figure 5: Signed-score stability matrix: rows are dataset and half-width, columns are tested masses. The gray cells are unavailable 2015 masses. This is a post-hoc diagnostic display, with one common color scale; it is not a reduced discovery search.

Here $\hat{\mu}_d^+$ is the independent nonnegative signal amplitude expressed as fitted signal events in that width's likelihood window. These are conditional fitted yields; they are not efficiency-corrected rates. The full-template yield and the original amplitude parameter A_d are also saved in the ledger. In that parameterization $A_d = \epsilon_d^2/10^{-8}$, used here simply as an independent template normalization for each dataset. All twelve 92 MeV fitted amplitudes are positive. Their local reference Z values alone do not demonstrate equal peak locations or a common physical coupling.

Dataset	w	Signed $Z(92)$	Local $p(92)$	$\hat{\mu}_d^+$ [events]	$\hat{A}_d$
2015	2.25	2.594	0.0047	1 147	7748.35
2015	2.4	2.594	0.0047	1 173	7857.10
2015	2.5	2.596	0.0047	1 188	7922.75
2015	2.6	2.691	0.0036	1 233	8196.43
2016	2.25	2.796	0.0026	16 252	913.25
2016	2.4	2.726	0.0032	16 182	902.53
2016	2.5	2.778	0.0027	16 662	924.29
2016	2.6	2.770	0.0028	16 703	924.58
2021	2.25	1.423	0.0774	7 585	165.47
2021	2.4	1.363	0.0864	7 436	159.77
2021	2.5	1.363	0.0864	7 436	159.77
2021	2.6	1.378	0.0841	7 650	163.06

Table 4: Conditional single-dataset local responses and fitted amplitudes at the inspected mass 92 MeV. Local tails use the source-standardized response model. The event yield is integrated over the actual fitted window; changing the window changes its contained fraction.

6 Null distribution of the coherence statistic

For comparison on identical support, the coherence statistic uses the post-hoc common interval 80–100 MeV. In each dataset and width, select the maximum signed reference score among sites with a positive raw fit. Exact ties choose the lower mass. If any dataset–width has no positive site, its complete diagnostic is assigned $T = W = +\infty$ and cannot count as coherent.

$$\bar{m} = \frac{\sum_{d,w} \hat{m}_{d,w} / \sigma_d^2(92)}{\sum_{d,w} 1 / \sigma_d^2(92)}, \quad T = \frac{1}{12} \sum_{d,w} \left[\frac{\hat{m}_{d,w} - \bar{m}}{\sigma_d(92)} \right]^2, \quad W = \frac{1}{3} \sum_d R_d^2.$$

Small T describes peaks clustered at one mass across both datasets and widths. W separately describes stability across width. The detector resolutions supply a dimensionless comparison scale; they are not claimed to be peak-estimator standard errors. These definitions were fixed in the continuation before extracting its observed peak table, but the region and scientific question arose after inspection. Their lower-tail probabilities are exploratory null diagnostics, not discovery significances, and are not multiplied together. Under the pinned null, 195 of 256 exact scans are at least

Statistic	Observed	Direct k/N	Direct p	Direct 95% interval	Gaussian p
T	2.9793	195/256	0.7626	[0.7047, 0.8126]	0.7387
W	9.4102	253/256	0.9883	[0.9661, 0.9976]	0.9858

Table 5: Inclusive lower-tail probabilities. T is the primary coherence diagnostic; W is a separate width-stability diagnostic. The region and statistics have not received a selection correction.

as tightly clustered as the observed twelve peaks. This is not unusually coherent. Likewise, 253 of 256 have width ranges at least as small. The Gaussian estimates agree within the direct intervals; increasing Gaussian statistics cannot remove source-model uncertainty. The paired Gaussian ensemble contains 63 ineligible scans, all retained in the denominator as noncoherent. No diagnostic tail is used to select a new mass region or nominal width.

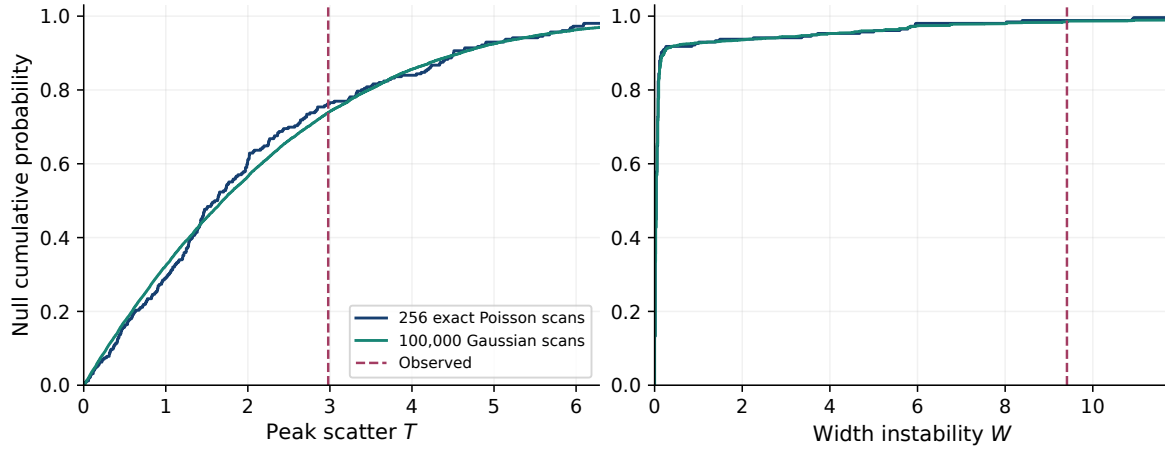


Figure 6: Inclusive lower-tail distributions from the 256 paired exact Poisson scans and 100,000 supplemental Gaussian-response scans. Each toy repeats the peak selection on the same grid and the same widths. Dashed lines mark observed values; smaller values indicate tighter coherence or greater stability. Ineligible toys remain in the denominator.

7 Validation

The numerical package contains 928 free-amplitude mass-width rows and 12 independent 92 MeV amplitude checks. The joint constrained likelihood was replayed at 20 representative mass-width configurations, including single-, two- and three-dataset regions, and checked against the factorized statistic and nested shared-coupling fit. All source Poisson spectra regenerate exactly from their recorded seeds, and all twelve observed 92 MeV signed roots reproduce the parent values exactly in the matching SciPy/NumPy environment.

The central independent-boundary mixture and the positive/zero-atom limits pass. Increasing quadrature order changes the checked tails by at most $1.6\text{e-}14$ relatively; lookup interpolation changes observed $\log p$ by at most $2.4\text{e-}07$. Joint covariance reconstruction differs by at most $7.6\text{e-}12$. Direct scalar evaluation reproduces the coherence metric to $4.4\text{e-}16$. Numerical checks establish implementation consistency, not physical source validity.

At each width's Gaussian 95th-percentile global threshold, the direct Poisson exceedance counts are 16, 17, 20 and 17 out of 256. These are above the nominal mean 12.8; all corresponding 95% binomial intervals still contain 0.05. This limited ensemble cannot rule out a modest response-tail mismatch. At the observed peaks, all Gaussian global estimates lie inside the direct intervals.

8 Discussion

The free-amplitude likelihood identifies a combined excess at 91–92 MeV throughout the four width choices. At the nominal width, the individual local reference significances at 92 MeV are 2.59, 2.80, and 1.42 for 2015, 2016, and 2021, respectively. The combined local significance is 3.583; scanning the full declared domain reduces it to 2.060 in the Gaussian-response approximation. Direct Poisson scans give a broader assessment of this tail, with seven exceedances among 256 scans.

The stable combined maximum does not extend to every individual regional maximum. In particular, a competing 2021 excursion overtakes the 93 MeV peak at the larger widths. The mass-coherence probability of 0.763 therefore provides no additional evidence for unusually aligned peaks. Both conclusions remain conditional on the fitted background source and the diagnostic

choices prompted by the observed spectra. The background source was fitted to the observed data and then frozen. Neither ensemble propagates source-estimation uncertainty, possible signal absorption during source construction, background-model selection, or unspecified correlated systematic uncertainties. The direct toys do refit the local GP predictions and profile the likelihood on each complete generated spectrum. The response fields approximate those operations by their stored derivatives. Agreement with 256 toys is a limited check, especially in sparse tails, not proof of precise Gaussian-tail calibration.

The 2015 support extension through 100 MeV, 2021 sample fraction, resolution functions, kernel states and signal templates remain inherited assumptions. No continuous-mass convergence claim is made: these results describe the 1 MeV grid. The parent study separately examined 0.5 versus 1 MeV spacing. The current extension does not choose a finer grid or smaller search domain after seeing a preferable result.

8.1 Reproducibility

The source bundle includes all pinned inputs needed by this continuation, copied likelihood code, analysis and plotting scripts, the protocol, numerical ledgers, saved null maxima and coherence statistics, validation records, and SHA-256 manifests. The released v5.8.4 study and output package were hashed before and after the continuation. Rebuild commands are in `README.md`; the original report is included unchanged and precedes this extension in the combined PDF.

References

- [1] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, arXiv:1007.1727.
- [2] E. Gross and O. Vitells, *Trial factors for the look elsewhere effect in high energy physics*, arXiv:1005.1891.
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- [4] HPS-GPR v5.8.4, *Independent combinations, mass spacing and blind-window comparisons*, pinned report and inputs, 18 September 2026.